

Exercise class 5

1. Discrete Fourier Transform, inverse transform
2. RC circuits, impedance
 - a) Derive the impedance formula for a series circuit $Z(\omega) = R + 1/(i\omega C)$. Discuss the low and high frequency behaviour.
 - b) Consider a harmonic periodic current and show that $|Z(\omega)|$ sets the amplitude, while $\arg Z(\omega)$ gives the phase shift of the voltage.
 - c) Show that for a parallel RC circuit the net impedance obeys the usual law for resistances.
3. Low pass filter (LPF): response to a box signal
 - a) Show that the real-time response function of the LPF is $f(t) = \frac{\sqrt{2\pi}}{RC} \Theta(t) e^{-t/(RC)}$.
 - b) Using the convolution theorem, derive the output signal for a box input signal (V_0 for $0 < t < T$ and 0 otherwise).